

One-sample and Two-sample Z and t Tests for a Mean and Mean difference

One-sample Z test

3-step test:

- (1) test statistic: $Z^* = \frac{(\bar{x} - \mu_0)}{\sigma/\sqrt{n}}$
- (2) critical value:
 - two sided: $Z_{\frac{\alpha}{2}}$
 - right sided: Z_{α}
 - left sided: $-Z_{\alpha}$
- (3) decision:
 - Two sided: If $|Z^*| > Z_{\frac{\alpha}{2}}$, then reject H_0
 - Right sided: If $Z^* > Z_{\alpha}$, then reject H_0
 - Left sided: If $Z^* < -Z_{\alpha}$, then reject H_0

p-value:

- Two sided: p-value = $2P(Z > |Z^*|)$
- Right sided: p-value = $P(Z > Z^*)$
- Left sided: p-value = $P(Z < Z^*)$

If p-value $< \alpha$, then reject H_0 .

(1- α) 100% CI for μ

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

One-sample t test

3-step test:

- (1) test statistic: $t^* = \frac{(\bar{x} - \mu_0)}{s/\sqrt{n}}$
- (2) critical value:
 - two sided: $t_{\frac{\alpha}{2}, n-1}$
 - right sided: $t_{\alpha, n-1}$
 - left sided: $-t_{\alpha, n-1}$
- (3) decision:
 - Two sided: If $|t^*| > t_{\frac{\alpha}{2}, n-1}$, then reject H_0
 - Right sided: If $t^* > t_{\alpha, n-1}$, then reject H_0
 - Left sided: If $t^* < -t_{\alpha, n-1}$, then reject H_0

p-value:

- Two sided: p-value = $2P(t_{n-1} > |t^*|)$
- Right sided: p-value = $P(t_{n-1} > t^*)$
- Left sided: p-value = $P(t_{n-1} < t^*)$

If p-value $< \alpha$, then reject H_0 .

(1- α) 100% CI for μ

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$$

Two-sample Z test

3-step test:

- (1) test statistic: $Z^* = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
- (2) critical value: same as one-sample Z test
- (3) decision: same as one-sample Z test

p-value: same as one-sample Z test

(1- α) 100% CI for $\mu_1 - \mu_2$

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Two-sample t test (assuming $\sigma_1 = \sigma_2$)

3-step test:

- (1) test statistic: $t^* = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$
where $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$
- (2) critical value: similar as one-sample t test, but change degree of freedom from $n - 1$ to $n_1 + n_2 - 2$
- (3) decision: similar as one-sample t test, but change degree of freedom from $n - 1$ to $n_1 + n_2 - 2$

p-value: similar as one-sample t test, but change degree of freedom from $n - 1$ to $n_1 + n_2 - 2$

(1- α) 100% CI for $\mu_1 - \mu_2$

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$